



Bleeding Disorders

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Bleeding Disorders due to defects in coagulation cascade

Inherited coagulation disorders

1. Hemophilia:

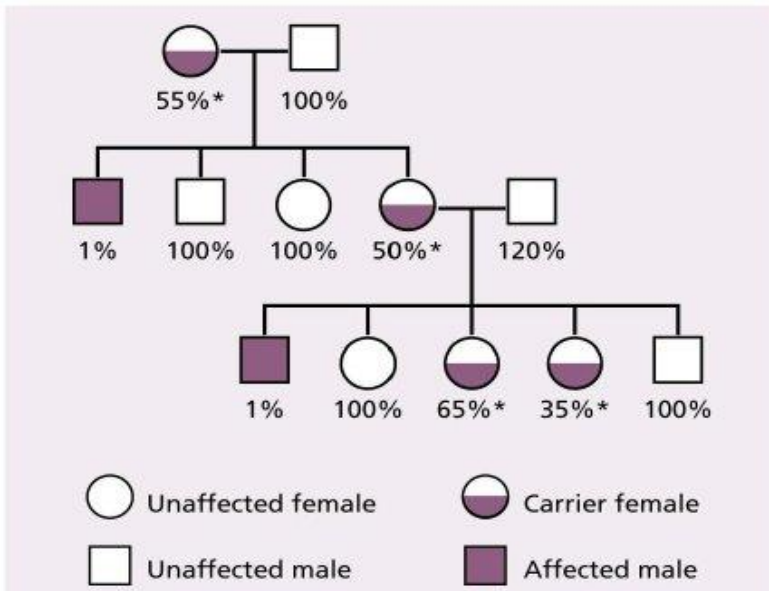
Hemophilia A (factor VIII deficiency)

Hemophilia B factor IX deficiency)

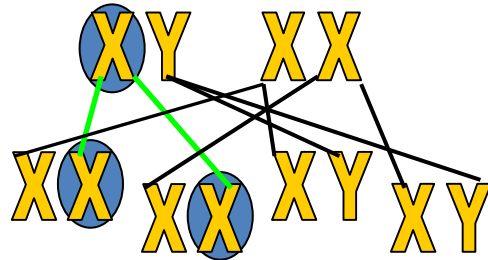
2. von Willebrand Disease (vWD)

Hemophilia

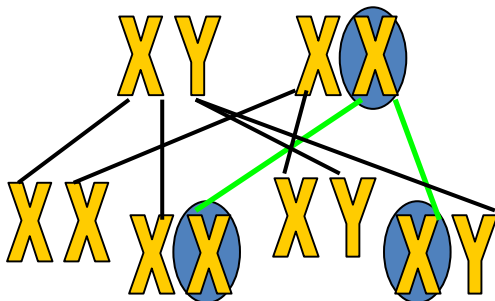
- Hemophilia is a rare disorder in which your blood doesn't clot normally because it lacks sufficient blood-clotting proteins (clotting factors).
- Hemophilia is an inherited bleeding disorder with a deficiency or lack of factor VIII (hemophilia A) or factor IX (hemophilia B).
- Hemophilia A and B are X-linked recessive disorders.
- Hemophilia is typically expressed in males and carried by females.



- Affected males
 - All daughters are carriers
 - No sons are affected



- Female carrier
 - 50% risk for carrier daughter
 - 50% risk for the affected son



Detection of Hemophilia

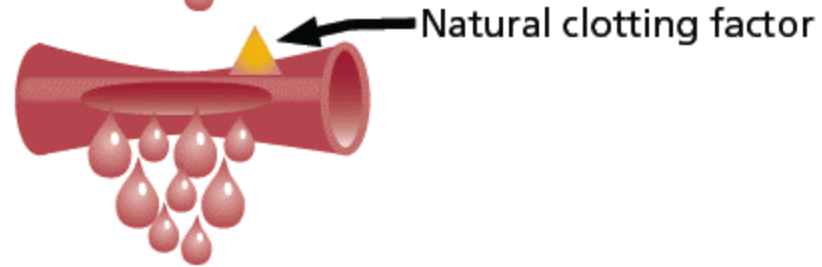
1. Family history
2. Symptoms:
 - Bruising
 - Bleeding with circumcision
 - Muscle, joint, or soft tissue bleeding
3. Hemostatic challenges (severe bleeding in case of:)
 - Surgery
 - Dental work
 - Trauma, accidents
4. Laboratory testing

Injury Occurs

Injury to blood vessel results in bleeding.



Vessel constricts and clotting factors are activated.

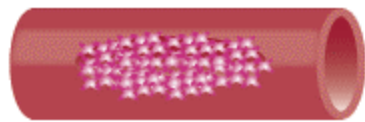


Normal

Natural clotting factor helps form a strong platelet plug.

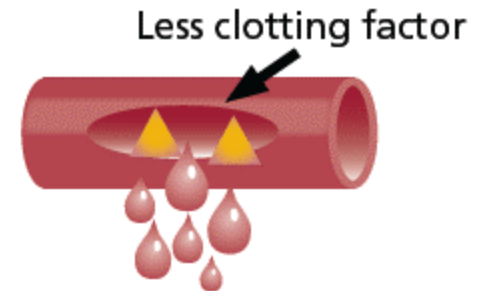


A stable fibrin mesh forms a sealed clot over the platelet plug to stop the bleeding.



Hemophilia

Lack of natural clotting factor means only a weak platelet plug can form.



Incomplete fibrin mesh allows bleeding to continue.



Types of Bleeds

1. Joint bleeding – hemarthrosis
2. Muscle hemorrhage
3. Soft tissue
4. Life threatening-bleeding
5. Other

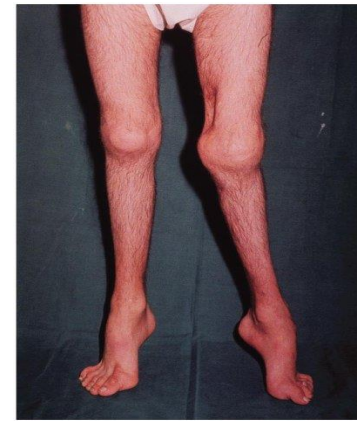
Joint or Muscle Bleeding

➤ Symptoms

1. Tingling or bubbling sensation
2. Stiffness
3. Warmth
4. Pain
5. Unusual limb position



(a)



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Life-Threatening Bleeding

- Head / Intracranial
 - Nausea, vomiting, headache, drowsiness, confusion, visual changes, loss of consciousness.
- Neck and Throat
 - Pain, swelling, difficulty breathing/ swallowing.
- Abdominal / GI
 - Pain, tenderness, swelling, blood in the stools.
- Iliopsoas Muscle
 - Back pain, abdominal pain, thigh tingling/ numbness, decreased hip range of motion.

Other Bleeding Episodes

1. Mouth bleeding
2. Nose bleeding
3. Scrapes and/or minor cuts
4. Menorrhagia

Complications of Bleeding

1. Flexion contractures
2. Joint arthritis / arthropathy
3. Chronic pain
4. Muscle atrophy
5. Compartment syndrome
6. Neurologic impairment

Degrees of Severity of Hemophilia

- Normal factor VIII or IX level has 50-150% activity
- Mild hemophilia
 - factor VIII or IX level = 6-50%
- Moderate hemophilia
 - factor VIII or IX level = 1-5%
- Severe hemophilia
 - factor VIII or IX level = <1%

Table 26.1 Correlation of coagulation factor activity and disease severity in haemophilia A or B.

Coagulation factor activity (percentage of normal)	Clinical manifestations
<1	Severe disease Frequent spontaneous bleeding into joints, muscles, internal organs from early life Joint deformity and crippling if not adequately prevented or treated
1-5	Moderate disease Bleeding after minor trauma Occasional spontaneous episodes
>5	Mild disease Bleeding only after significant trauma, surgery

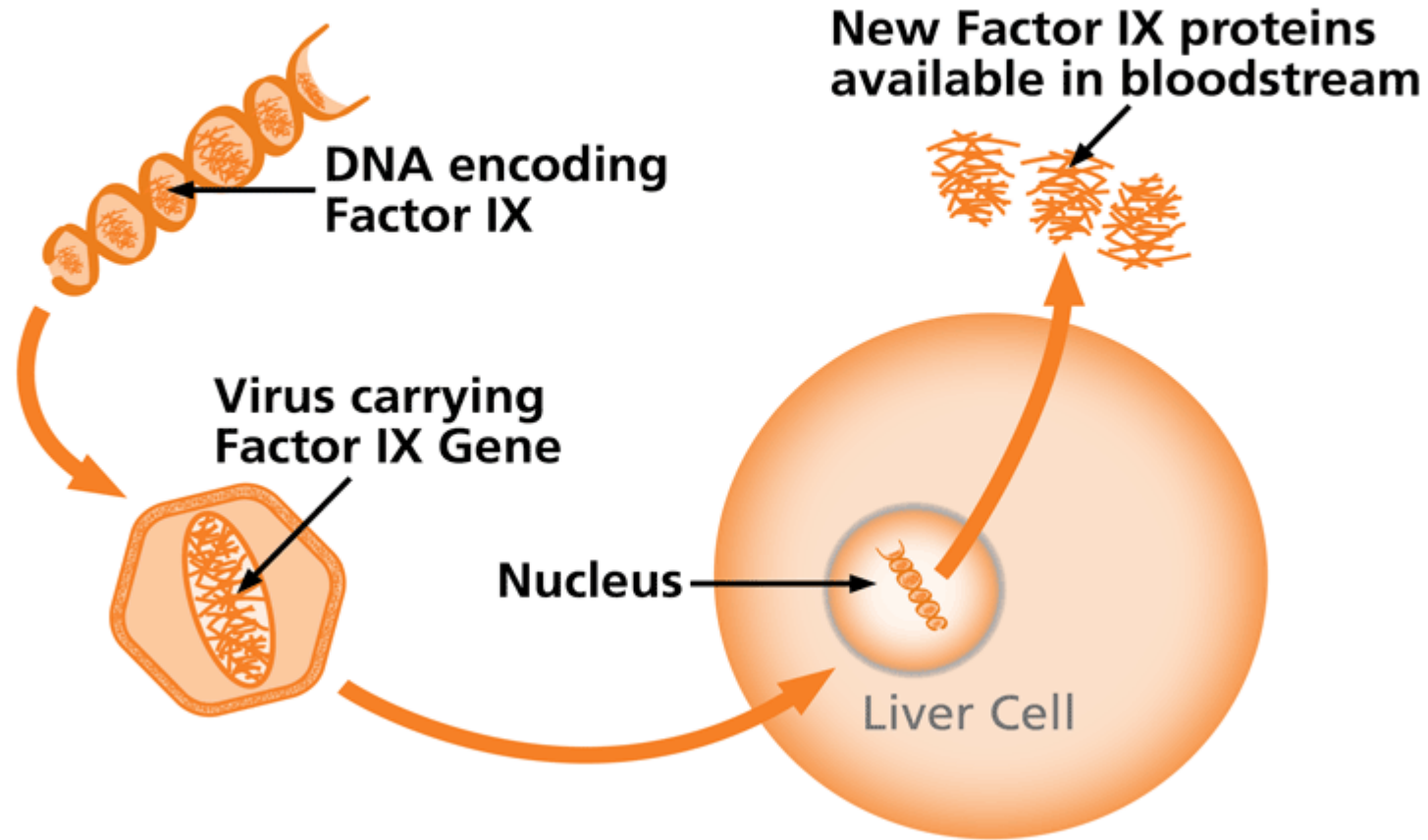
Treatment of hemophilia

- Affected child should be monitored for life sports activity and at school.
- Emergency plans should be developed to control bleeding as fast as possible.
- Coagulation factors should be administered on time.

Disorder	Treatment of choice	Alternative
Hemophilia A	Desmopressin	Recombinant Factor VIII
Hemophilia B	Recombinant Factor IX	

- Blood transfusion in case of severe hemorrhage.
- Fresh frozen plasma or cryoprecipitate is another option to obtain coagulation factors.

Gene Therapy – Type B



Nature Outlook: Haemophilia. The Search for Genetic Treatments for Haemophilia Is Proving Difficult. Dec 16, 2014.
Image adapted from: Duke University: Pharmacology 150. How Gene Therapy Works.

von Willebrand Disease

Von Willebrand Factor:

- The gene for the vWF subunit is located on chromosome 12p.
- Homomultimeric glycoprotein ranging in size from 600,000 to 20 million daltons
- The carbohydrate component is estimated to add 10-15% to the molecular mass.
- Synthesized in endothelial cells and megakaryocytes and stored in Weibel-Palade bodies and platelet alpha granules, respectively.
- vWF is initially formed in the ER as a pre-pro VWF molecule, which would assemble into homomultimeric protein after glycosylation, dimerization, and multimerization in the Golgi organelle and storage places in the cells.

- vWD is inherited in an autosomal dominant pattern.
- Approximately 80% of cases are classified as type 1 disease, which is a quantitative deficiency in vWF and presents with mild bleeding symptoms;
- 15% to 20% of patients are classified as having type 2 disease.
- Type 2A disease results from a qualitative deficiency of vWF in which there is an abundance of the small but absent of the hemostatically effective large multimers, thus causing defective platelet adhesion at the site of vessel injury.

- In Type 2B vWD, platelet function is enhanced due to an increased affinity of vWF binding to glycoprotein Ib. There also may be associated thrombocytopenia.
- Type 2N disease is associated with decreased FVIII binding and mimics hemophilia.
- Finally, type 3 disease is a rare autosomal recessive disorder characterized by the absence of vWF and severe bleeding.

Table 26.3 Classification of von Willebrand disease.

Type 1 Quantitative partial deficiency

Type 2 Functional abnormality

Type 3 Complete deficiency

Secondary classification of type 2 VWD

Subtype	Platelet-associated function	Factor VIII binding capacity	High MW VWF multimers
2A	Decreased	Normal	Absent
2B	Increased affinity for GPIb	Normal	Usually reduced/absent
2M	Decreased	Normal	Normal
2N	Normal	Reduced	Normal

GPIb, glycoprotein Ib; MW, molecular weight; VWD, von Willebrand disease; VWF, von Willebrand factor.

Function of vWF:

- Serves as the carrier protein for factor VIII (probably factor VIII and vWF are brought together in storage granules).
- Serves as the ligand that binds to glycoprotein Ib receptor on platelets to initiate platelet adhesion to damaged blood vessel walls.
- vWF needs to be activated to bind to GP 1b receptor on platelets (Ristocetin, high shear force, collagen, etc).

Differential Lab diagnosis

Table 26.2 Main clinical and laboratory findings in haemophilia A, factor IX deficiency (haemophilia B, Christmas disease) and von Willebrand disease.

	Haemophilia A	Factor IX deficiency	von Willebrand disease
Inheritance	Sex-linked	Sex-linked	Dominant (incomplete)
Main sites of haemorrhage	Muscle, joints, post-trauma or postoperative	Muscle, joints, post-trauma or postoperative	Mucous membranes, skin cuts, post-trauma or postoperative
Platelet count	Normal	Normal	Normal
PFA-100	Normal	Normal	Prolonged
Prothrombin time	Normal	Normal	Normal
Partial thromboplastin time	Prolonged	Prolonged	Prolonged or normal
Factor VIII	Low	Normal	May be moderately reduced
Factor IX	Normal	Low	Normal
VWF	Normal	Normal	Low or abnormal function (Table 26.3)
Ristocetin-induced platelet aggregation	Normal	Normal	Impaired

VWF, von Willebrand factor.

Hemophilia A is more common than B

Type	Inheritance	Cause	Frequency
Hemophilia A	X linked recessive	Factor VIII	80%
Hemophilia B	X linked recessive	Factor IX	15%
Von Willebrand disease	Autosomal dominant	von Willebrand factor	5%

	Hemophilia A	Hemophilia B	Von Willebrand Disease
Inheritance	X linked	X linked	Autosomal dominant
Factor deficiency	VIII	IX	VWF
Bleeding site(s)	Muscle,joint Surgical	Muscle ,joint	Mucous Skin
Prothrombin time	Normal	Normal	Normal
Activated PTT	Prolonged	Prolonged	Prolonged
Bleeding time	Normal	Normal	Prolonged or normal
Platelet aggregation	Normal	Normal	Normal

Disseminated Intravascular Coagulation (DIC)

- Also known as **consumptive coagulopathy**.
- Is a generalized pathological activation of the hemostasis mechanism that happens in response to a variety of diseases characterized by the widespread formation of blood clots in the small blood vessels throughout the body.
- DIC involves all hemostatic systems: platelets, coagulation factors, coagulation control pathways, and fibrinolysis.
- The formation of small blood clots (fibrin microthrombi) inside the blood vessels throughout the body partially occlude small vessels.

- In DIC, there is a consumption of platelets, coagulation factors, coagulation control proteins (protein C, protein S, and Anti-thrombin), and fibrinolytic enzymes.
- Fibrin/ fibrinogen degradation products (FDPs, traditionally called fibrin split products, FSPs), including D-dimer, become elevated and interfere with normal fibrin formation.
- DIC may be acute and uncompensated, with deficiencies of multiple hemostasis components, or chronic, with normal or even elevated clotting factor levels.
- Acute DIC is often fatal and requires immediate medical intervention.

- In chronic DIC, liver coagulation factor production and BM platelet production compensate for increased consumption.
- Although DIC is a thrombotic process, the thrombi that form are small and ineffective, so systemic hemorrhage is often the first or most apparent sign.
- As the small clots consume all the available coagulation proteins and platelets, normal coagulation is disrupted and abnormal bleeding occurs from the skin, the digestive tract, the respiratory tract, and surgical wounds.
- The small clots also disrupt normal blood flow to organs (such as the kidneys), which may malfunction as a result (organ failure).
- It is common in the critically ill and may participate in the development of multiple organ failure, which may lead to death.

DIC Causes

- Any disorder that contributes to hemostatic molecules or promotes their endogenous secretion may cause DIC.
- The more acutely ill patients, the more dangerous symptoms.
- Chronic DIC may be associated with vascular tumors, tissue necrosis, liver and renal disease, chronic inflammation, and prosthetic devices.
- Acute DIC is seen as associated with obstetric emergencies, intravascular hemolysis, septicemia, viremia, burns, acute inflammation, cardiac disorder, and crush injuries.

Pathophysiology

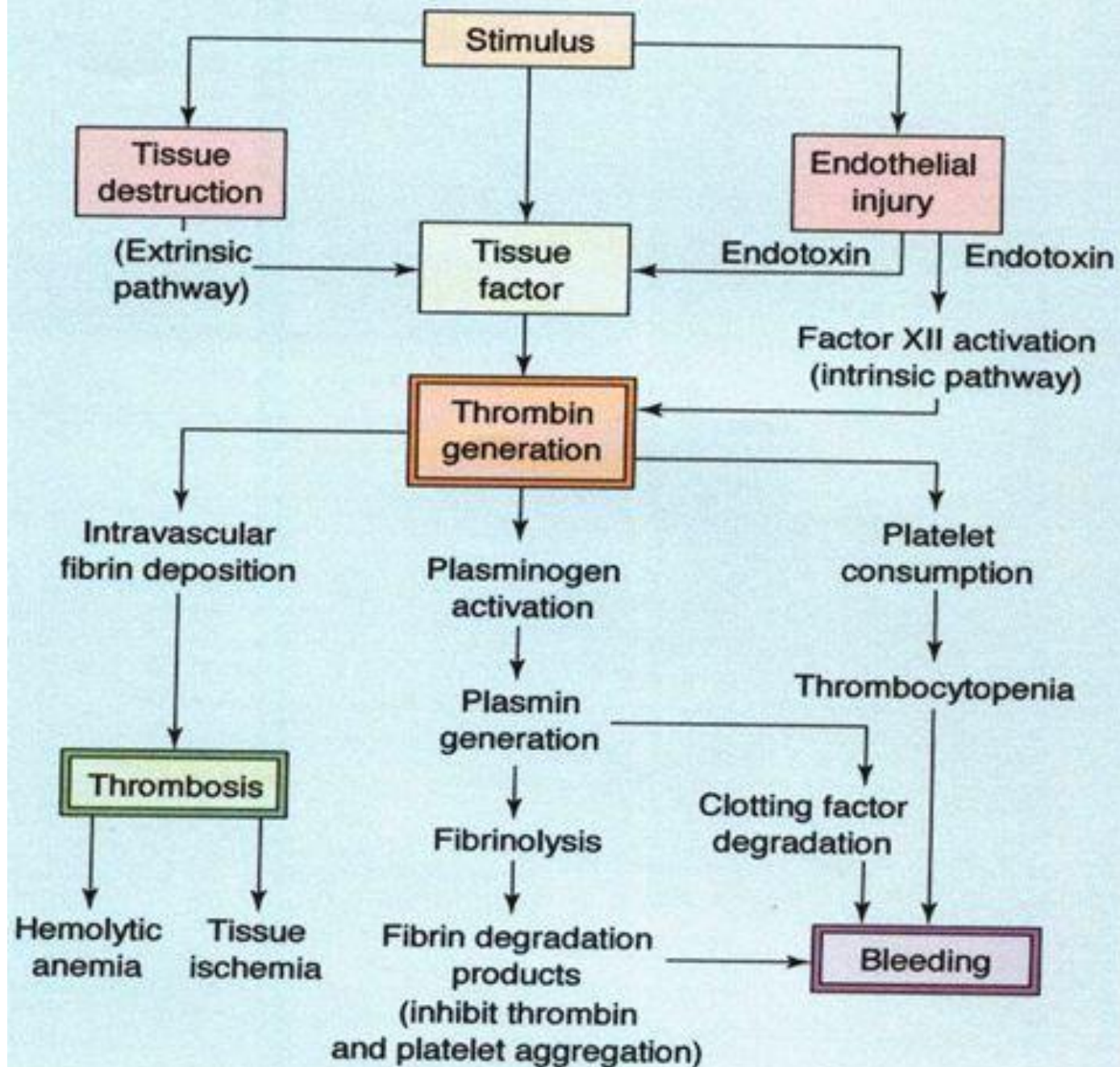
- Under normal homeostatic conditions, the body is maintained in a finely tuned balance of coagulation and fibrinolysis.
- The activation of the coagulation cascade yields thrombin that converts fibrinogen to fibrin to form the stable fibrin clot (fibrin polymer).
- The fibrinolytic system then functions to break down fibrinogen and fibrin.
- The breakdown of fibrinogen and fibrin results in polypeptides called fibrin degradation products (FDPs) or fibrin split products (FSPs).
- In DIC, the processes of coagulation and fibrinolysis lose control, and the result is widespread clotting with resultant bleeding.

- Regardless of the triggering event of DIC, once initiated, the pathophysiology of DIC is similar in all conditions.
- One critical mediator of DIC is the release of a transmembrane glycoprotein called tissue factor (TF).
- TF is present on the surface of many cell types (including endothelial cells, macrophages, and monocytes) and is not normally in contact with the general circulation, but is exposed to the circulation after vascular damage.

- Excess circulating thrombin results from the excess activation of the coagulation cascade.
- The excess thrombin cleaves fibrinogen, which ultimately leaves behind multiple fibrin clots in the circulation.
- These excess clots trap platelets to become larger clots, which leads to microvascular and macrovascular thrombosis.
- Soluble fibrin monomers, fibrin polymer, and cross-linked fibrin activate plasminogen.

- Normally, the active form of plasminogen, plasmin, acts locally to digest only the solid fibrin clot to which it is bound.
- In DIC, plasmin circulates in the plasma and digests all forms of fibrin.
- Consequently, FDPs (X, Y, E, D, and D-dimer) are detectable.
- Control of the coagulation pathway (Coagulation inhibitors) are also consumed as protein C, protein S, and Antithrombin.
- Decreased inhibitor levels will permit more clotting so that a feedback system develops in which increased clotting leads to more clotting.

- At the same time, thrombocytopenia occurs because of the entrapment and consumption of platelets.
- Finally, the combination of thrombin activation, circulating plasmin, loss of control, and thrombocytopenia contribute to the overall hemorrhagic outcome of DIC.



Signs and symptoms

- DICs are frequently masked by the symptoms of the underlying disorders.
- Thrombosis in the microvasculature of major organs may produce symptoms of organ failure, such as renal failure, adult respiratory distress syndrome, and central nervous system manifestation.
- Skin, bones, and bone marrow necrosis may be seen.
- The affected person is often acutely ill and shocked with widespread hemorrhage (common bleeding sites are mouth, nose, and venipuncture sites), extensive bruising, and gangrene.

Laboratory Diagnosis

- DIC is a clinical diagnosis that requires laboratory confirmation.
- The initial laboratory profile includes a platelet count, blood film examination, PT, PTT, D-dimer, and fibrinogen assay.
- DIC is discovered on laboratory testing.
- The diagnosis relies on the hemostasis laboratory (DIC profile).

- Diagnosis is usually suggested by the following conditions:

Severe cases with hemorrhage:

- ▶ The PT and PTT are usually very prolonged.
 - ▶ The fibrinogen level is markedly reduced.
-
- High levels of fibrin degradation products, including D-dimer, are found owing to the intense fibrinolytic activity stimulated by the presence of fibrin in the circulation.
 - There is severe thrombocytopenia.
 - The blood film may show fragmented red blood cells (schistocytes).

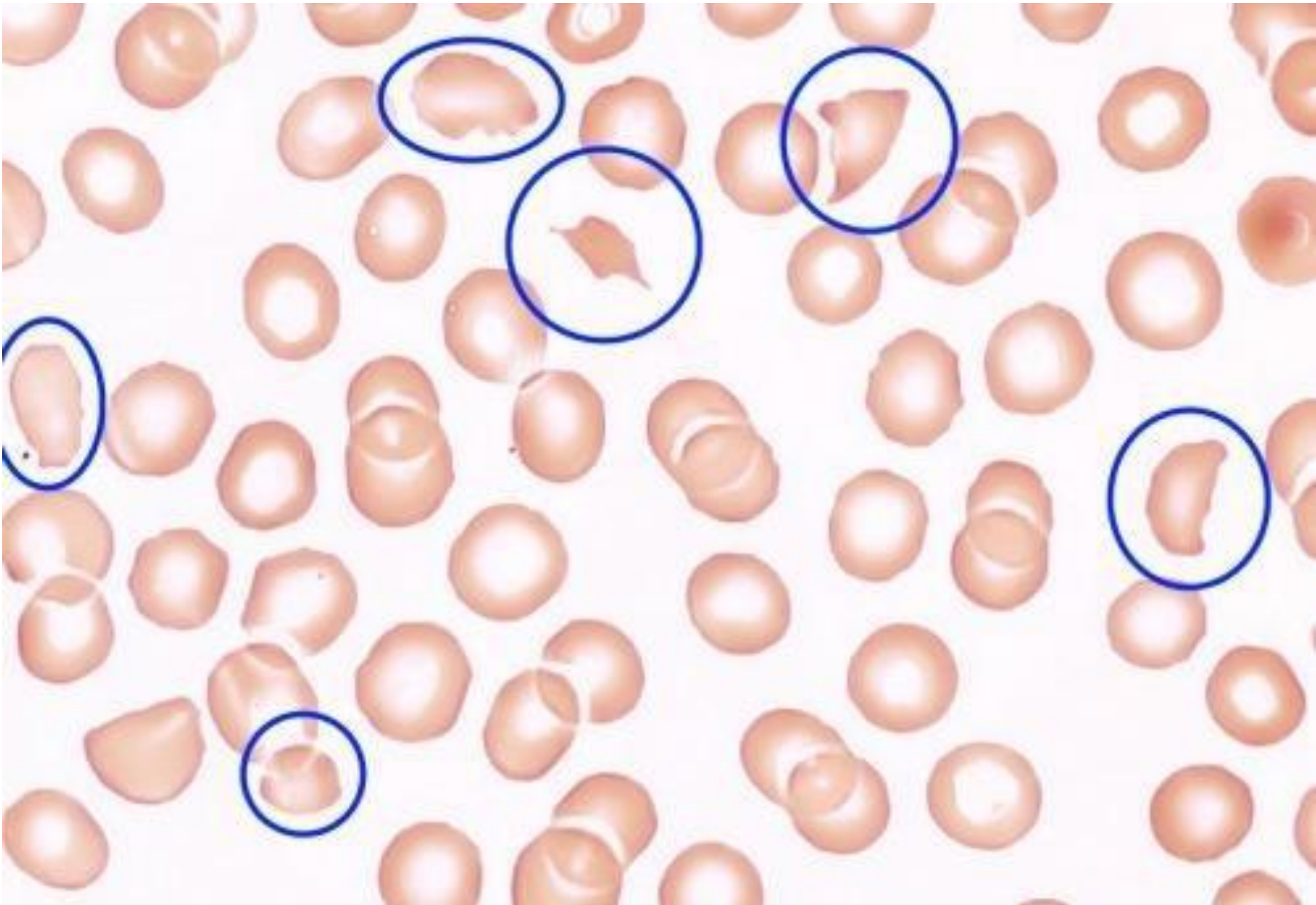
Mild cases without bleeding:

- ▶ There is increased synthesis of coagulation factors and platelets.
 - ▶ PT, PTT, and platelet counts are normal.
 - ▶ Fibrin degradation products are raised.
-
- Fibrinogen levels may decrease in DIC, but because fibrinogen is an acute-phase reactant that increases inflammation, **the fibrinogen alone provides little information.**
 - An elevated D-dimer level is essential to the diagnosis.
 - In 85% of DIC cases, D-dimer levels reach 10,000 to 20,000 ng/mL.
 - Protein C, protein S, and antithrombin typically are consumed in DIC, and their assay may contribute to the diagnosis.

- D-dimer also may be elevated in inflammatory conditions, localized thrombosis, or renal diseases in the absence of DIC.
- The PT, PTT, platelet count, and blood film examination must be performed with the D-dimer to accurately the cause of elevated D-dimer.
- A normal D-dimer assay result rules out DIC.
- Definitive diagnosis depends on the result of:
 1. Thrombocytopenia.
 2. Prolongation of PT and PTT.
 3. A low fibrinogen concentration.
 4. Increased levels of fibrin degradation.

TABLE 39-15 Anticipated Results of Disseminated Intravascular Coagulation (DIC) Primary Laboratory Profile

Test	Reference Interval	Value in DIC
Platelet count	150,000–450,000/ μ L	<150,000/ μ L
Prothrombin time	11–14 sec	>14 sec
Partial thromboplastin time	25–35 sec	>35 sec
D-dimer	0–240 ng/mL	>240 ng/mL, often 10,000 to 20,000 ng/mL
Fibrinogen	220–498 mg/dL	<220 mg/dL, often higher, because fibrinogen is an acute phase reactant



Blood smear with typical schistocytes in DIC marked in blue